NAME	:	
STUDENT NO.	:	
GROUP	:	

PART A: CIRCLE THE CORRECT ANSWER.

1 - Select the CORRECT statement about to describe a queue.

A. A queue is referred to as a LIFO data structure.
B. Elements are added at one end and deleted from the other end.
C. `enqueue()` is the method used to access the first element in a queue.
D. The last element inserted into a queue will be the first element to be removed.

2 - Which data structure is the most appropriate to be used to represent a service counter.

A. Stack
B. Array List
C. Queue
D. Binary Search Tree

3 - This is a data structure that implements the first-in-first-out (FIFO) protocol.

A. Sequential list
B. Linked list
C. Stack
D. Queue

4 - Which of the following is TRUE about queues?

i) A queue is referred to as a LIFO data structure because the last item inserted is the first item removed.
ii) It is a type of list
iii) The method to remove data from a queue is known as `enqueue()`.

A. i, ii and iii.
B. i and iii only.
C. ii and iii only.
D. ii only.

PART B

Observe the following `Product` and `Queue` ADTs:

```
public class Product
{
      private String type; //shortSleeve, shortPants, booties
      private String color; //blue, pink
      private char size; //S, M, L
      private double price;

      //normal constructor
      //getters
      //printers
}

public class Queue
{
      public void enqueue(Object elem) { }
      public Object dequeue() { }
      public Boolean isEmpty() { }
      //definition of other methods
}
```

Write a Java program segment using the above ADTs:

(a) Store **TEN (10)** products into the queue; `prodQueue`.
(b) Copy all products from `prodQueue` and store them into **THREE (3)** different queues (`shortSleeve`, `shortPants` and `booties`). **All products in the `prodQueue` must remain in the original order.
(c) Calculate and display the total price of all products from each queue.
(d) Display all the types and prices of the products of size '*M*' and colour "*blue*".